

ROBERTO GUAGNELLI GARCÍA

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Mechanical Engineering Student | Mechanical Design, CAD & Aircraft Integration

EDUCATION

Tecnológico de Monterrey, Campus Estado de México Expected Jul 2027
B.S. in Mechanical Engineering; Concentration in Aerospace Engineering
Academic average: 92/100 (3.7/4.0 GPA)

WORK EXPERIENCE

GE Aerospace | Designer Intern, Product Definition Aug 2026 – Present
RISE Open Fan Program

- Develop electrical harness routing and mounting bracket CAD models in Siemens NX; review interferences and apply design for manufacturability (DFM) and maintainability criteria to component integration.
- Prepare engineering drawings defining component geometry and design requirements for the RISE Open Fan program.

GE Aerospace | Lean Summer Experience Intern Jun 2026 – Jul 2026
Integrated Performance & Operability (IP&O)

- Implemented visual planning, dashboards and agreed review practices with subsection managers and program leads; reduced average weekly meeting duration by 20%, as timed by meeting leaders.
- Defined and tracked key performance indicators (KPIs) and supported cloud database migration to improve performance visibility across management and engineering teams.

ENGINEERING PROJECTS & LEADERSHIP

Borregos CEM AeroDesign | Tecnológico de Monterrey 2024 – Present
Engineering Coordinator / Design Captain Aug 2024 – Present

- Lead a five-person design team and coordinate 40+ members across design, structures, electronics and R&D; developed Wingull (2025) and Latías (2026) from conceptual design through flight testing.
- Personally modeled 300+ CAD parts per aircraft within 400+ component assemblies; evaluated mass, center of gravity, stability, materials and motor/propeller combinations to guide design decisions.
- Performed steady-state and transient computational fluid dynamics (CFD) studies of airfoils and vortex formation to assess aerodynamic performance and inform stability evaluation; used drag coefficients in takeoff models to estimate payload capability at different density altitudes.
- Designed removable joints for Wingull's 4.2 m wing to enable breakdown into components under 1.2 m; supported an 18 kg distributed-mass wing test with measured tip deflection within 8% of predictions.
- Redesigned a 3D-printed PETG bottle retainer for Latías after insertion-force and fracture issues; revised geometry completed 50 insertion/removal cycles without observed cracks or permanent deformation.
- Integrated Latías' twin-engine configuration, which flew with 6 kg payload at 7 kg empty mass; Wingull flew with 4 kg payload at 6 kg empty mass.

Manufacturing Coordinator | Hyperion 2024

- Coordinated eight members in aircraft fabrication, assembly and quality control; designed and manufactured a welded-aluminum nose landing gear with a spring, travel stop and fork to accommodate limited propeller clearance.
- Supported six competition flights carrying 3 kg payload at 4 kg empty mass; team placed 2nd overall and 3rd in design report at SAE Aero Design México 2024.

TECHNICAL SKILLS

CAD & Analysis: Siemens NX, SolidWorks, Fusion 360, ANSYS Fluent, XFLR5; steady-state and transient CFD.

Programming & Data: Python, MATLAB, Excel, Power BI.

Design & Testing: Engineering drawings, interference checks, DFM, maintainability, propulsion selection, structural and impact testing.

Manufacturing: 3D printing, carbon-fiber fabrication, laser cutting, aluminum welding, tube and sheet bending.

Certification: Certified SOLIDWORKS Professional (CSWP), perfect exam score.

Languages: Spanish (native), English (C2; Cambridge and IELTS), German (A2).